

# 3. User's aur System's Point of View of Operating System

Computer Science Fundamentals | Module 1: OS Architecture | CS\_Fundamental-00

**Reference to Previous Concepts:** Concept 1.2 mein OS ko intermediary aur Concept 2.1 mein Abstract View dekha gaya tha. Yeh topic **Image 3** par aadharit hai jisme OS ko do mukhya perspectives se dekha jata hai: **User View** aur **System View (Resource Manager)**.

## 3.1 User View (End-User Perspective)

- **Primary Goal: Convenience, Usability & Responsiveness:** User ke liye OS ka matlab hai aasan aur bina rukawat computer chalana.
- **Interface Modes:** GUI (Windows, icons), CLI (Terminal commands), aur APIs (Developers ke liye).
- **User Focus:** Running applications, managing files, configuring settings, and interacting with peripherals.

## 3.2 System View (Resource Manager Perspective)

Image 3 handwritten note: "**Resource Manager**" (marked very important).

- **Primary Goal: Efficiency, Control & Resource Allocation:** System perspective se OS ek Resource Allocator hai. Hardware limited hai (CPU cores, RAM, I/O bandwidth). OS ka kaam hai fair aur collision-free distribution karna.
- **Core Components:** Kernel, Device Drivers, Memory Management, Process Management, File Systems, Networking Protocols.
- **Emphasis:** Efficiency, fair resource allocation, security aur stability.

**Fig 3.1: Dual Perspectives of Operating System**

Reference: Image 3 [User View vs System View / Resource Manager]

### 1. USER VIEW (End-User)

#### Primary Goal: Convenience & Usability

- Platform for executing applications
- Interfaces: GUI, CLI, APIs
- Focus: Running apps, files, settings
- User concerns: Responsiveness, ease of use

#### Example: User Perspective

User writes code:

```
a = b + c;
```

User only cares about output correctness and fast execution!

### 2. SYSTEM VIEW (Resource Manager)

#### Primary Goal: Efficiency & Control

- OS as a master Resource Allocator
- Controls CPU, Memory, I/O devices
- Enforces Security, Stability & Isolation
- Target audience: SysAdmins, Engineers

#### Example: System Perspective

Inside OS & Hardware:

1. Memory allocated for a, b, c
2. Values loaded in registers (R1, R2)
3. CPU executes ADD, result written to RAM
4. Protection prevents illegal memory access

Fig 3.1: Dual Perspectives of Operating System

### 3.3 Numerical / Computational Example & Step-by-Step Thought Process: $a = b + c$

Image 3 handwritten note:  $a = b + c$ . Aaiye iska complete thought process samjhein:

**Scenario: User writes** `int b = 10; int c = 20; int a = b + c;`

**1. User View Thought Process:** User chahta hai ki `10 + 20` add hokar turant `30` mil jaye. User ko underlying hardware address se matlab nahi hai.

**2. Translators Role (Ref Concept 2.2):** Compiler ne iss instruction ko machine code mein toda:

- `LOAD R1, [Addr_B]` (Value 10 in Register R1)
- `LOAD R2, [Addr_C]` (Value 20 in Register R2)
- `ADD R3, R1, R2` (ALU performs addition: 30)
- `STORE [Addr_A], R3` (Value 30 stored in RAM address A)

**3. System View (Resource Manager) Thought Process:**

- Process Allocation:* OS iss code ko PID deta hai aur isolated RAM allot karta hai.
- Memory Protection:* MMU check karta hai ki process apni boundary ke bahar na jaye.
- CPU Scheduling:* OS process ko CPU cycle allot karta hai aur context switch ke samay register state save karta hai.
- I/O Signaling:* Final output screen driver ke zariye user ko deliver hota hai.

### 3.4 User View vs System View Comparison Table

| Feature                | User View (End-User)            | System View (Resource Manager)                        |
|------------------------|---------------------------------|-------------------------------------------------------|
| Main Objective         | Convenience & Usability         | Efficiency & Resource Optimization                    |
| Interaction Mode       | GUI, CLI, Desktop Apps          | System Calls, Interrupts, Drivers                     |
| Focus Area             | App execution & output response | Memory Allocation, CPU Scheduling, Security           |
| Perspective on $a=b+c$ | "Sirf 30 answer aana chahiye."  | "RAM allocate hui, Registers load hue, MMU verified!" |